

Lec4

Introduction to CSS

CSS (Cascading Style Sheets) is a style sheet language used to control the **appearance and layout** of web pages. It works together with HTML.

- **HTML** provides the structure/content of a webpage.
- **CSS** provides the design and styling of that content.

For example, HTML creates a heading, while CSS can change its color, size, font, and position.

Why css is Used?

To add colors to web pages.

To change fonts and text styles.

To control spacing and layout.

To create responsive designs for different devices.

To improve the overall look and user experience.

Example

HTML

```
<h1>Hello World</h1>  
<p>Welcome to CSS.</p>
```

CSS

```
h1 {  
  color: blue;  
  text-align: center;  
}
```

```
p {  
  color: green;  
  font-size: 18px;  
}
```

CSS Syntax

A CSS rule consists of:

- **Selector:** Selects the HTML element.
- **Property:** The style attribute to change.
- **Value:** The value assigned to the property.

```
selector {  
  property: value;  
}
```

Types of CSS

Inline CSS

Internal CSS

External CSS

Inline CSS

Use inline styles to apply styling to a specific element. Note that this is not optimal. Placing style rules in a `<style>` tag or external CSS file is encouraged in order to maintain a distinction between content and presentation.

Inline styles override any CSS in a `<style>` tag or external style sheet. While this can be useful in some circumstances, this fact more often than not reduces a project's maintainability.

The styles in the following example apply directly to the elements to which they are attached.

```
<h1 style="color: green; text-decoration: underline;">Hello world!</h1>
```

```
<p style="font-size: 25px; font-family: 'Trebuchet MS';">I ♥ CSS</p>
```

Inline styles are generally the safest way to ensure rendering compatibility across various email clients, programs and devices, but can be time-consuming to write and a bit challenging to manage.

Internal CSS

CSS enclosed in `<style></style>` tags within an HTML document functions like an external stylesheet, except that it lives in the HTML document it styles instead of in a separate file, and therefore can only be applied to the document in which it lives. Note that this element must be inside the `<head>` element for HTML validation (though it will work in all current browsers if placed in body).

Example

```
<head>
<style>
h1 {
color: green;
text-decoration: underline;
}
p {
font-size: 25px;
font-family: 'Trebuchet MS', sans-serif;
}
</style>
</head>
<body>
<h1>Hello world!</h1>
<p>I ♥ CSS</p>
</body>
```

External CSS

An external CSS stylesheet can be applied to any number of HTML documents by placing a `<link>` element in each HTML document.

The attribute `rel` of the `<link>` tag has to be set to "stylesheet", and the `href` attribute to the relative or absolute path to the stylesheet. While using relative URL paths is generally considered good practice, absolute paths can be used, too. In HTML5 the `type` attribute can be omitted.

It is recommended that the `<link>` tag be placed in the HTML file's `<head>` tag so that the styles are loaded before the elements they style. Otherwise, users will see a flash of unstyled content.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8" />
    <link rel="stylesheet" type="text/css" href="style.css">
  </head>
  <body>
    <h1>Hello world!</h1>
    <p>I ☐ CSS</p>
  </body>
</html>
```

```
h1 {
  color: green;
  text-decoration: underline;
}
p {
  font-size: 25px;
  font-family: 'Trebuchet MS', sans-serif;
}
```

CSS Selectors

Selectors are used to select HTML elements that you want to style with CSS.

Types of Selectors

Tag Selector

ID Selector

Class Selector

Universal Selector

Descendant Selector

Child Selector

Tag Selector

Selects all elements of a specific HTML tag.

HTML

```
<h1>Heading</h1>
```

```
<p>Paragraph</p>
```

CSS

```
h1 {  
  color: blue;  
}
```

```
p {  
  color: green;  
}
```

ID Selector

Selects an element with a specific `id`.

HTML

```
<h1 id="title">Welcome</h1>
```

CSS

```
#title {  
  color: red;  
}
```

Class Selector

Selects all elements with a specific class.

HTML

```
<p class="text">First Paragraph</p>  
<p class="text">Second Paragraph</p>
```

CSS

```
.text {  
  color: purple;  
}
```


Universal Selector

Selects all elements.

```
* {  
  margin: 0;  
  padding: 0;  
}
```

Group Selector

Applies the same style to multiple elements.

```
h1, h2, p {  
  color: blue;  
}
```

Descendant Selector

Selects elements inside another element.

HTML

```
<div>
  <p>Paragraph</p>
</div>
```

CSS

```
div p {
  color: green;
}
```

Child Selector

Selects direct children only.

HTML

```
<div>
  <p>Direct Child</p>
  <section>
    <p>Not Direct Child</p>
  </section>
</div>
```

CSS

```
div > p {
  color: red;
}
```

Properties

Color

Background

Text

Font

Task

Style an HTML page using external CSS